MSG No. 133.470 P_I4_2/nbc [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `4'22'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,4,5,6,18,19,23,24]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,3,7,8,17,20,21,22]`
3	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[9,12,13,14,26,27,31,32]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[10,11,15,16,25,28,29,30]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4'2m'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[1,4,5,6,26,27,31,32]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,3,7,8,25,28,29,30]`
3	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[9,12,13,14,18,19,23,24]`
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[10,11,15,16,17,20,21,22]`

Table 3: Wyckoff site: `4c`, site symmetry: `4'/m'..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,6,10,11,18,19,25,30]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,9,14,17,22,26,27]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[4,5,15,16,23,24,28,29]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[7,8,12,13,20,21,31,32]`

Table 4: Wyckoff site: `4d`, site symmetry: `m'.m'm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[1,6,7,8,25,30,31,32]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,4,5,26,27,28,29]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[9,14,15,16,17,22,23,24]`
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[10,11,12,13,18,19,20,21]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2'/m'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,9,23,31]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[2,10,21,29]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,11,20,28]`
4	$[0, 0, 0]$	`[4,12,19,27]`
5	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,13,18,26]`
6	$\left[\frac{1}{2}, 0, 0\right]$	`[6,14,24,32]`
7	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[7,15,17,25]`

8	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[8,16,22,30]`

Table 6: Wyckoff site: `8f`, site symmetry: `4'..`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,6,18,19]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[2,3,17,22]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[4,5,23,24]`
4	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[7,8,20,21]`
5	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[9,14,26,27]`
6	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[10,11,25,30]`
7	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[12,13,31,32]`
8	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[15,16,28,29]`

Table 7: Wyckoff site: `8g`, site symmetry: `2.m'm'`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[1,6,31,32]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[2,3,28,29]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	`[4,5,26,27]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[7,8,25,30]`
5	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	`[9,14,23,24]`
6	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[10,11,20,21]`
7	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[12,13,18,19]`
8	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[15,16,17,22]`

Table 8: Wyckoff site: `8h`, site symmetry: `m'.2m'`

No.	position	mapping
1	$\left[x,\ x,\ \frac{3}{4}\right]$	`[1,7,30,32]`
2	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	`[2,5,27,28]`
3	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[3,4,26,29]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[6,8,25,31]`
5	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[9,15,22,24]`
6	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	`[10,13,19,20]`
7	$\left[-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[11,12,18,21]`
8	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[14,16,17,23]`

Table 9: Wyckoff site: `16i`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[1,23]`
2	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[2,21]`
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[3,20]`

continued ...

Table 9

No.	position	mapping
4	$[x, -x, 0]$	[4,19]
5	$[\frac{1}{2} - x, x + \frac{1}{2}, 0]$	[5,18]
6	$[\frac{1}{2} - x, -x, 0]$	[6,24]
7	$[x + \frac{1}{2}, x, \frac{1}{2}]$	[7,17]
8	$[-x, \frac{1}{2} - x, \frac{1}{2}]$	[8,22]
9	$[-x, \frac{1}{2} - x, 0]$	[9,31]
10	$[x, -x, \frac{1}{2}]$	[10,29]
11	$[\frac{1}{2} - x, x + \frac{1}{2}, \frac{1}{2}]$	[11,28]
12	$[-x, x, 0]$	[12,27]
13	$[x + \frac{1}{2}, \frac{1}{2} - x, 0]$	[13,26]
14	$[x + \frac{1}{2}, x, 0]$	[14,32]
15	$[\frac{1}{2} - x, -x, \frac{1}{2}]$	[15,25]
16	$[x, x + \frac{1}{2}, \frac{1}{2}]$	[16,30]

Table 10: Wyckoff site: 16j, site symmetry: .2.

No.	position	mapping
1	$[x, \frac{1}{4}, 0]$	[1,4]
2	$[\frac{1}{4}, x, \frac{1}{2}]$	[2,7]
3	$[\frac{1}{4}, \frac{1}{2} - x, \frac{1}{2}]$	[3,8]
4	$[\frac{1}{2} - x, \frac{1}{4}, 0]$	[5,6]
5	$[-x, \frac{3}{4}, 0]$	[9,12]
6	$[\frac{3}{4}, -x, \frac{1}{2}]$	[10,15]
7	$[\frac{3}{4}, x + \frac{1}{2}, \frac{1}{2}]$	[11,16]
8	$[x + \frac{1}{2}, \frac{3}{4}, 0]$	[13,14]
9	$[x + \frac{1}{2}, \frac{3}{4}, \frac{1}{2}]$	[17,20]
10	$[\frac{3}{4}, x + \frac{1}{2}, 0]$	[18,23]
11	$[\frac{3}{4}, -x, 0]$	[19,24]
12	$[-x, \frac{3}{4}, \frac{1}{2}]$	[21,22]
13	$[\frac{1}{2} - x, \frac{1}{4}, \frac{1}{2}]$	[25,28]
14	$[\frac{1}{4}, \frac{1}{2} - x, 0]$	[26,31]
15	$[\frac{1}{4}, x, 0]$	[27,32]
16	$[x, \frac{1}{4}, \frac{1}{2}]$	[29,30]

Table 11: Wyckoff site: 16k, site symmetry: m'..

No.	position	mapping
1	$[x, y, \frac{3}{4}]$	[1,30]
2	$[\frac{1}{2} - y, x, \frac{1}{4}]$	[2,27]
3	$[y, \frac{1}{2} - x, \frac{1}{4}]$	[3,26]
4	$[x, \frac{1}{2} - y, \frac{1}{4}]$	[4,29]
5	$[\frac{1}{2} - x, y, \frac{1}{4}]$	[5,28]
6	$[\frac{1}{2} - x, \frac{1}{2} - y, \frac{3}{4}]$	[6,25]
7	$[y, x, \frac{3}{4}]$	[7,32]

continued ...

Table 11

No.	position	mapping
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[8,31]
9	$\left[-x,\ -y,\ \frac{1}{4}\right]$	[9,22]
10	$\left[y+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	[10,19]
11	$\left[-y,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[11,18]
12	$\left[-x,\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[12,21]
13	$\left[x+\frac{1}{2},\ -y,\ \frac{3}{4}\right]$	[13,20]
14	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[14,17]
15	$\left[-y,\ -x,\ \frac{1}{4}\right]$	[15,24]
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[16,23]

Table 12: Wyckoff site: 16l, site symmetry: ..m'

No.	position	mapping
1	$[x,\ x,\ z]$	[1,32]
2	$\left[\frac{1}{2}-x,\ x,\ z+\frac{1}{2}\right]$	[2,28]
3	$\left[x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[3,29]
4	$\left[x,\ \frac{1}{2}-x,\ -z\right]$	[4,26]
5	$\left[\frac{1}{2}-x,\ x,\ -z\right]$	[5,27]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z\right]$	[6,31]
7	$\left[x,\ x,\ \frac{1}{2}-z\right]$	[7,30]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[8,25]
9	$[-x,\ -x,\ -z]$	[9,24]
10	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	[10,20]
11	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[11,21]
12	$\left[-x,\ x+\frac{1}{2},\ z\right]$	[12,18]
13	$\left[x+\frac{1}{2},\ -x,\ z\right]$	[13,19]
14	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	[14,23]
15	$\left[-x,\ -x,\ z+\frac{1}{2}\right]$	[15,22]
16	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[16,17]

Table 13: Wyckoff site: 32m, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$	[2]
3	$\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[3]
4	$\left[x,\ \frac{1}{2}-y,\ -z\right]$	[4]
5	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	[5]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[6]
7	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[7]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[8]
9	$[-x,\ -y,\ -z]$	[9]
10	$\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	[10]
11	$\left[-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[11]

continued ...

Table 13

No.	position	mapping
12	$\left[-x,\, y+\frac{1}{2},\, z\right]$	[12]
13	$\left[x+\frac{1}{2},\, -y,\, z\right]$	[13]
14	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[14]
15	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[16]
17	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[17]
18	$\left[-y,\, x+\frac{1}{2},\, z\right]$	[18]
19	$\left[y+\frac{1}{2},\, -x,\, z\right]$	[19]
20	$\left[x+\frac{1}{2},\, -y,\, \frac{1}{2}-z\right]$	[20]
21	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[21]
22	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[22]
23	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[23]
24	$[-y,\, -x,\, -z]$	[24]
25	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[25]
26	$\left[y,\, \frac{1}{2}-x,\, -z\right]$	[26]
27	$\left[\frac{1}{2}-y,\, x,\, -z\right]$	[27]
28	$\left[\frac{1}{2}-x,\, y,\, z+\frac{1}{2}\right]$	[28]
29	$\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[29]
30	$\left[x,\, y,\, \frac{1}{2}-z\right]$	[30]
31	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[31]
32	$[y,\, x,\, z]$	[32]